FIITJEE **JEE (Advanced)-2018**

ALL INDIA TEST SERIES

FULL TEST – V

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 231**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into three sections: **Section-A, Section-C & Section-D**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. ***Section-A (01 – 03, 24 – 26, 47 – 49)*** *contains 9 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

   Section-A (04 – 08, 27 – 31, 50 – 54) *contains 15 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 marks*** *for wrong answer.*

   Partial Marks **+1** for each correct option provided no incorrect options is selected.

   Section-A (09 – 10, 32 – 33, 55 – 56) *contains 3 paragraphs. Based upon paragraph, 2 multiple choice questions have to be answered. Each question has* ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer. There is no negative marking.*

2. ***Section-C (11 – 20, 34 – 43, 57 – 66)*** *contains 30 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

3. ***Section-D (21 – 23, 44 – 46, 67 – 69)*** *contains 9 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

PART – I (Physics), PART – II (Chemistry), PART – III (Mathematics):

(SECTION – D)

For questions **21 to 23, 44 to 46, 67 to 69.**
Numerical answer type questions with answer XXXXX. XX

If answer is 348.4 / 251.37 / 213

Correct Method :

0	0	3	4	8	.	4	0
0	0	2	5	1	.	3	7
0	0	2	1	3	.	0	0

Wrong Method :

	3	4	8		.	4	
3	4	8			.		4
		3	4	8	.		4
	3		4	8	.	4	
	2		5	1	.	3	7
		2	1	3	.		
		2	1	3	.	0	
		2	1	3	.		0
		3	4	8	.	4	0
		2	5	1	.	3	7
		2	1	3	.	0	0

Physics

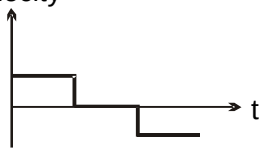
PART – I

SECTION – A
(One Options Correct Type)

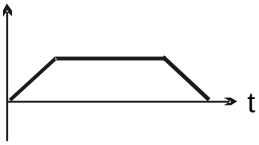
This section contains **3 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D) out of which **ONLY ONE** option is correct.

1. A boy is pushing a box on a horizontal floor from a position of rest to rest while moving on a straight line. Consider the three phases of motion. Floor is rough with a small friction coefficient.
(i) Initially a constant hard push on the box to get it moving and attain a maximum velocity.
(ii) Mild push to keep the box moving with constant velocity.
(iii) To pull back the box to bring it to stop with same retardation.
Which of the following graphs is **CORRECT?**

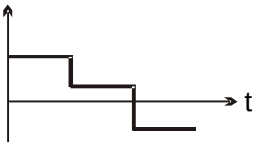
(A) Velocity



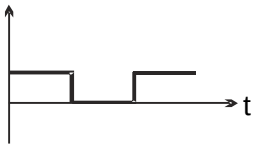
(B) Acceleration



(C) Force exerted by worker



(D) Friction force



2. A light ray incident along vector $2\hat{i}+4\hat{j}+\sqrt{5}\,\hat{k}$ strikes on the x-z plane from medium of refractive index $\sqrt{3}$ and enters into medium II of refractive index is μ_2. The value of μ_2 for which the value of angle of refraction becomes 90°, is

(A) $\frac{4\sqrt{3}}{5}$ (B) $\frac{3\sqrt{3}}{5}$

(C) $\frac{2\sqrt{3}}{5}$ (D) $\frac{\sqrt{3}}{5}$

Space for Rough work

3. According to Bohr's theory, the radius of the nth orbit of an atom of atomic number Z is proportional to

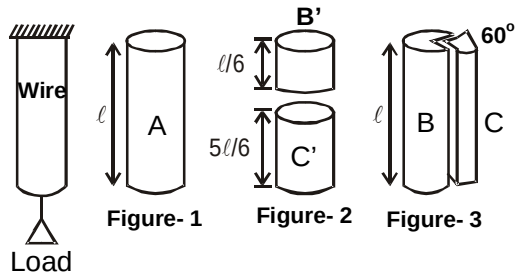
(A) $\frac{n^2}{z^2}$ (B) $\frac{n^2}{z}$

(C) $\frac{n}{z}$ (D) n^2z^2

(One or More than one correct type)

This section contains **FIVE** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

4. A light wire of length l (figure - 1) is cut into two pieces in two different ways as shown in (figure – 2 & 3). Different pieces can be arranged in placed of wire as shown and a load can be placed on the massless hanger. Choose the correct statement(s).



(A) The load required to break the wire B' is 6 times that required to break B
(B) The stress required to break the wire B and C is same but to break B and B' is different
(C) The stress required to break C and C' is same
(D) The load required to break A and B' is same, but different for B and B'

5. Two blocks, A and B, of same masses, are resting in equilibrium on an inclined plane having inclination with horizontal $=\alpha(>0)$. The blocks are touching and exerting non zero normal force on each other with block B higher than A. Coefficient of static friction of A with incline = 1.2 and of B = 0.8. If motions not imminent

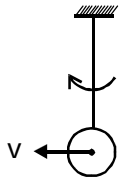
(A) $\alpha < 30^o$ (B) $(\text{Friction})_A > (\text{Friction})_B$

(C) $\alpha < 45^o$ (D) $(\text{Friction})_A = (\text{Friction})_B$

Space for Rough work

6. Two balls A & B of mass m_1 and m_2 are kept on a horizontal smooth surface. A is given a velocity towards B so that they perform head on collision
(A) If $m_1 = m_2$ and collision is elastic A stops and B moves with the velocity of A after collision
(B) Impulses on A and B during collision are same in magnitude whether it is elastic or inelastic collision
(C) Impulses on A and B during collision are equal in magnitude only if $m_1 = m_2$
(D) If $m_1 > m_2$, though impulses are same in magnitude velocity of A is still in the same direction.

7. A disc of radius a is rigidly attached at its circumference to a rod of length 3a and the combination suspended vertically from the other end of the rod. It is swinging in the plane of the disc such that the centre of disc has velocity v. If mass of disc and rod is m



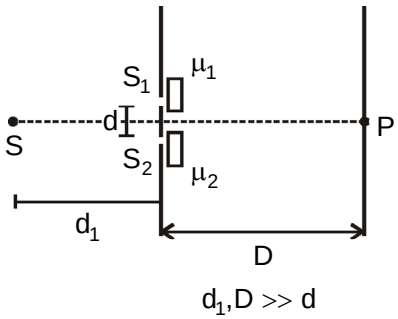
(A) Kinetic energy of the disc is $\frac{1}{2}mv^2$

(B) Kinetic energy of the disc $\frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{ma^2}{2}\right)\left(\frac{v}{a}\right)^2$

(C) Kinetic energy of the system of disc and rod is $\left[\frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{ma^2}{2}\right)\left(\frac{v}{4a}\right)^2\right] + \frac{1}{2}\left[\frac{m(3a)^2}{3}\left(\frac{v}{4a}\right)^2\right]$

(D) Kinetic energy of the system of disc and rod is $\frac{1}{2}\left[\frac{m(3a)^2}{3} + \left(\frac{ma^2}{2} + m(4a)^2\right)\right]\left(\frac{v}{4a}\right)^2$

8. In young's experiment, the upper slit is covered by a thin glass plate of refractive index $\frac{4}{3}$ and of thickness 9λ, where λ is the wavelength of light used in the experiment. The lower slit is also covered by another glass plate of thickness $2D$ & refractive index $\frac{3}{2}$, as shown in figure. If I_0 is the intensity at point P due to slits S_1 & S_2 each, then:



$d_1, D >> d$

(A) Intensity at point P is $4I_0$
(B) Two fringes have been shifted in upward direction after insertion of both the glass plates together.
(C) Optical path difference between the waves from S_1 & S_2 at point P is 2λ.
(D) If the source S is shifted upwards by a small distance d_2 then the fringe originally at P after inserting the plates, shifts downwards by $D\left(\frac{d_2}{d_1}\right)$

Space for Rough work

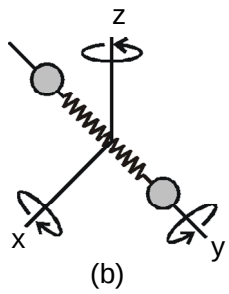
(Paragraph Type)

This section contains **ONE** paragraph. Based on the paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Paragraph for Question Nos. 9 and 10

Let us consider a diatomic gas whose molecules have the shape of a dumbbell. In this model, the center of mass of the molecule can translate in the x, y, and z directions. In addition, the molecule can rotate about three mutually perpendicular axes. We can neglect the rotation about the y-axis because the moment of inertia I_y and the rotational energy $\frac{1}{2}I_y\omega^2$ about this axis are negligible compared with those associated with the x and z axes. (If the two atoms are taken to be point masses, then I_y is identically zero). Thus, there are five degrees of freedom: three associated with the translational motion and two associated with the rotational motion. Because each degree of freedom contributes, on the average, $\frac{1}{2}k_BT$ of energy per molecule, the total internal energy for a system of N molecules is:

$$E_{int} = 3N\left(\frac{1}{2}k_BT\right) + 2N\left(\frac{1}{2}k_BT\right) = \frac{5}{2}Nk_BT = \frac{5}{2}nRT \quad .....(i)$$



(b)

We can use this result to find the molar specific heat at constant volume:

$$C_v = \frac{1}{n}\frac{dE_{int}}{dT} = \frac{1}{n}\frac{d}{dT}\left(\frac{5}{2}nRT\right) = \frac{5}{2}R \quad .....(ii)$$

From equation (i) and (ii), we find that

$$C_P = C_V + R = \frac{7}{2}R$$

$$\gamma = \frac{C_P}{C_V} = \frac{7/2R}{5/2R} = \frac{7}{5} = 1.40$$

In the vibratory model, the two atoms are joined by an imaginary spring. The vibrational motion adds two more degrees of freedom, which correspond to the kinetic energy and the potential energy associated with vibrations along the length of the molecule about its centre of mass.

9. The root mean square angular velocity of a diatomic molecule (with each atom of mass m and interatomic distance a) is given by:-

(A) $\sqrt{\frac{4k_BT}{ma^2}}$ (B) $\sqrt{\frac{2k_BT}{ma^2}}$

(C) $\sqrt{\frac{k_BT}{ma^2}}$ (D) $\sqrt{\frac{k_BT}{2ma^2}}$

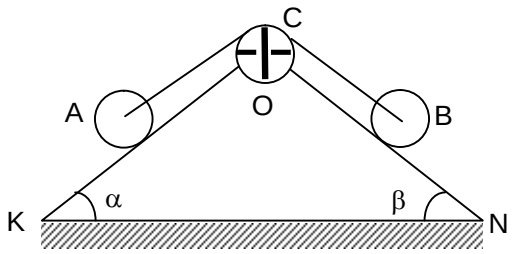
Space for Rough work

10. A diatomic molecule is moving without rotation or vibration with velocity v_{rms} such that it is oriented along x-axis. It strikes a wall in yz-plane while moving in +ve x direction. The spring constant can be assumed to be K and time of collision is negligible. After all collision are over:-
(A) The molecule is moving along – ve x-direction and oscillating about its centre of mass
(B) The molecule is moving along – ve x-direction, but not oscillating about its centre of mass
(C) The molecule is oscillating about it centre of mass but not moving at all.
(D) The molecule is neither rotating nor moving at all.

SECTION – C
(Single digit integer type)

This section contains **TEN** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

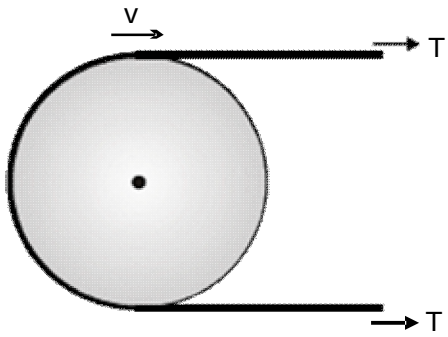
11. A wheel A is connected to a second wheel B by mean of inextensible string, passing over a pulley C. Which rotates about a fixed horizontal axle O, as shown in figure. The system is released from rest. The wheel A rolls down the inclined plane OK thus pulling up the wheel B which rolls along the inclined plane ON. Determine the velocity (in m/s) of the axle of the wheel A, when it has travelled a distance s = 3.5 m down the slope. Both wheels and the pulley are assumed to be homogeneous disks of identical weight and radius. Neglect the weight of the string. $\left[\text{Take } \alpha = 53^o \text{ and } \beta = 37^o\right]$



12. A ball is projected from a point on an inclined plane horizontally. It takes $\frac{1}{\sqrt{2}}$ seconds to strike the plane. If the incline plane makes an angle of 30° with horizontal, then find the range of the ball on incline plane in meters. $\left[g = 10 \text{ m/s}^2\right]$

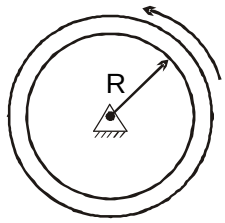
Space for Rough work

13. A flexible drive belt runs over a frictionless flywheel (see figure). The mass per unit length of the drive belt is 1 kg/m, and the tension in the drive belt is 10N. The speed of the drive belt is 2 m/s. The whole system is located on a horizontal plane. If the normal force (in N) exerted by the belt on the flywheel is N_f fill the value of $\frac{2}{3}N_f$ in OMR sheet.

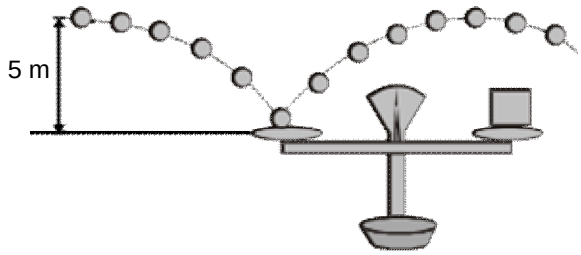


14. An elastic cord having an upstretched length l, force constant k and mass per unit length m_o is stretched around the drum of radius $r(2\pi r > l)$. Determine the speed of the cord, due to rotation of the drum, which will allow the cord to loosen its contact with the drum. Express value of $\frac{v^2}{11}\left(\text{in } m^2/s^2\right)$

$\left(\text{for the given data } m_0 = 40 \text{ g/cm}, k = 100 \text{ N/m}, \pi = \frac{22}{7}, l = 400 \text{ cm}, r = 70 \text{ cm}\right)$.



15. A stream of glass beads, each with a mass of 15 gram, comes out of a horizontal tube at a rate of 100 per second. The beads fall a distance of 5 m to a balance pan and bounce back to their original height. How much mass (in kg) must be placed in the other pan of the balance to keep the pointer at zero?

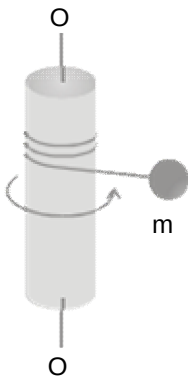


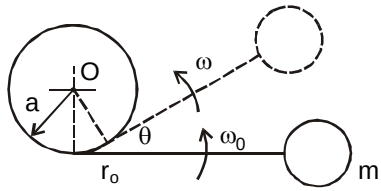
Space for Rough work

16. A block placed on a smooth horizontal floor is connected to a springs as shown. Initially an external force of 100 N keeps it stretched by 1 cm beyond natural length in equilibrium. This force is now removed and another force F is applied on this block, which slowly moves it from this position to a position where the spring is finally compressed by 3 cm. Find the work done by this force F (in joules).

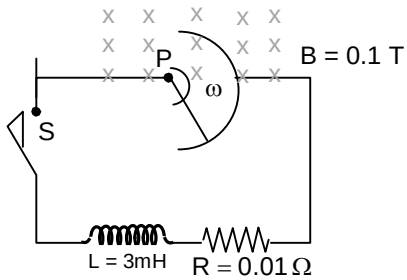


17. The small particle of mass m is given an initial high velocity in the horizontal plane and winds its cord around the fixed vertical shaft of radius 1 m. All motion occurs essentially in the horizontal plane. If the angular velocity of the cord is 0.8 rad/s when the distance from the particle to the tangency point is 5m., determine the angular velocity $\omega(\text{in rad/s})$ of the cord after it has turned through an angle 1 rad.





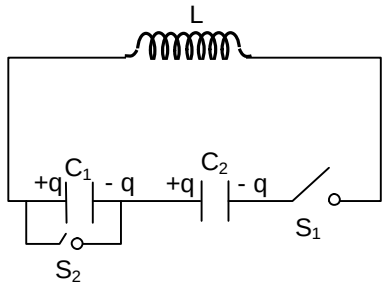
18. The circuit shows a resistance $R = 0.01\,\Omega$ and inductance L = 3 mH connected to a conducting rod PQ of length $l = 2\text{m}$ which can side on a perfectly conducting circular arc of radius l with its centre at P. Assume that friction & gravity are absent and a constant uniform magnetic field B = 0.1 T exists as shown in the figure. At t = 0, the circuit is switched on and simultaneously an external torque is applied on the rod so that it rotates about P with a constant angular velocity $\omega = 2$ rad/sec. Find the magnitude of this torque (in N-m) at t = (0.3 In 2) second



Space for Rough work

19. Two identical equi-concave lenses made of glass of reflected index 1.5 placed in contact has power P. When a liquid of refractive index μ is filled in the gap between the concave lenses, the power becomes of initial value (P/3). The value of μ is $\frac{K}{3}$. Find the value of K.

20. Two capacitors $C_1 = C_2 = \frac{1}{\pi^2} \times 10^{-2}$F and inductor L = 2×10^{-2}H are connected in series as shown in the figure. Initially charge on each capacitors are $4\sqrt{3}$ μC. At t = 0 switch S_1 is closed and t = 1/400 sec, switch S_2 is also closed. The maximum charge on capacitor C_2 during LC oscillation is $k\sqrt{3}$ μC. Find k.

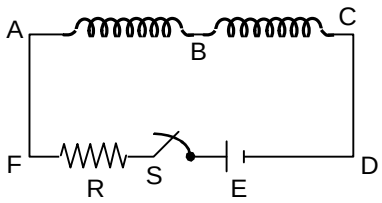


SECTION – D
(Numerical Based XXXXX.XX answer Type)

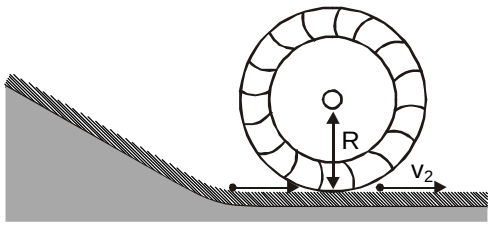
This section contains **3 questions**. Each question, when worked out will result in numerical answer Type with answer xxxxx.xx.

21. Two particles A and B located at points $\left(0,-10\sqrt{3}\right)$ and $(0,0)$ in xy plane. They start moving simultaneously at time t = 0 with constant velocities $\vec{V}_A = 5\hat{i}$ m/s and $\vec{V}_B = -5\sqrt{3}\,\hat{j}$ m/s, respectively. Time when they are closest to each other is found to be K/2 second. Find K. All distance are given in meter.

22. Two ideal solenoids of same dimensions. One is air cored with 600 turns while other is Aluminium is cored with 200 turns (relative permeability of Aluminium is 3), are connected in a circuit as shown in the figure. The switch S is closed at t = 0. Find the ratio of potential difference across air-cored solenoid to that of Aluminium cored solenoid at any time t.



23. Water flowing along an open channel drives an undershot waterwheel of radius 2 m (figure). The water approaches the wheel with a speed of 5.0 m/s and leaves with a speed of 2.5 m/s; the amount of water passing by is 200 kg per second. At what rate does the water deliver angular momentum to the wheel (in J)?



Undershot water wheel

Space for Rough work

Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **3 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D) out of which **ONLY ONE** option is correct.

24. $C_2H_5{\cdot}OC_2H_5 \longrightarrow A + B$

$\downarrow Fe^{2+}$

$\downarrow KCNS$

red color

Which is not true for above reactions?
(A) A & B are explosives
(B) If A & B are present in ether, then prior to its use, A & B must be removed from ether
(C) A & B can not be removed by adding conc H_2SO_4
(D) When A & B are treated with Fe^{2+} followed by KCNS, no redox reaction is involved.

25. If 1 litre aq. solution of H_2SO_4 (10% by weight) with density 1.1 gm/ml, find the pH of solution.
(A) $-\log 2.244$ (B) $-\log 1.96$
(C) $-\log 1.86$ (D) $-\log 3.87$

26. Which of the following sequence is correct for solubility in H_2O?
(1) $BeSO_4 > MgSO_4 > CaSO_4 > BaSO_4$
(2) $Be(OH)_2 > Mg(OH)_2 > Ca(OH)_2 > Ba(OH)_2$
(3) $LiOH < NaOH < KOH < RbOH$
(4) $Li_2CO_3 < Na_2CO_3 < K_2CO_3 < Rb_2CO_3$
(A) 1, 3, 4 (B) 1, 2, 4
(C) 2, 4 (D) 1, 3, 4

Space for Rough work

(One or More than one correct type)

This section contains **FIVE** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

27. $Ag_{(s)}\left|AgBr_{(s)}\right|Br^{-}(0.01M)\,||\,Cl^{-}(0.2M)\,|\,AgCl_{(s)}\,|\,Ag_{(s)}$ if K_{sp} of AgBr & AgCl at 25°C are 10^{-12} & 10^{-10} respectively find which is/are true for above cell at 25°C (log 2 = 0.3)

(A) As $E^{o}_{Ag^{+}/Ag}$ is not given so E_{cell} can not be determined

(B) E_{cell} is 0.0413 volt

(C) $\Delta G < 0$

(D) $E^{o} = 0$

28. $PhNO_2 \xrightarrow[NH_4Cl]{Zn} A \xrightarrow[Tollen's\,Reagent]{\substack{AgNO_3\\NH_4OH}} B$ (organic compound). Which is/are incorrect?

(A) A gives Tollen's test

(B) A shows tautomerism

(C) A is aniline

(D) A forms complex with Tollen's reagent but not gives silver mirror

29. Which of the following is/are incorrect option(s)?

(A) $HOCl < HClO_2 < HClO_3 < HClO_4$ (acidic strength)

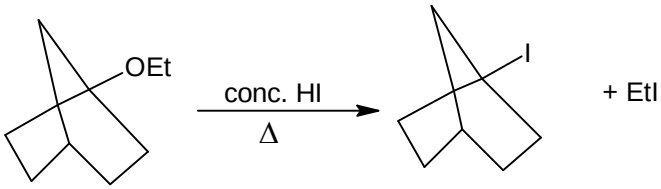
(B) $HOCl > HClO_2 > HClO_3 > HClO_4$ (thermal stability)

(C) $NH_3 > PH_3 > AsH_3$ (Basic strength)

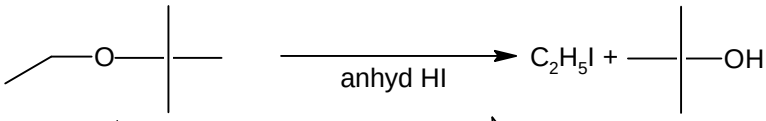
(D) $NH_3 > BiH_3 > SbH_3 > AsH_3 > PH_3$ (Boiling point)

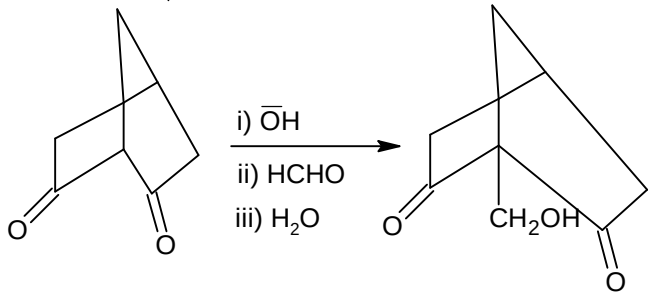
Space for Rough work

30. Which of the following is/are correct reaction?

(A) 

(B) $PhCH_2OMe \xrightarrow[\Delta]{conc.HBr} PhCH_2Br + MeBr$

(C) 

(D) 

31. In which of the following case(s) at least one bond angle is less than 90°?

(A) ClF_3 (B) SF_4
(C) BrF_5 (D) P_4

(Paragraph Type)

This section contains **ONE** paragraph. Based on the paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Paragraph for Question Nos. 32 and 33

Co-ordination complexes can show stereoisomerism square planar complexes of Ma_2b_2 type can show geometrical isomerism but not optical & octahedral complexes of $Ma_2b_2C_2$, Ma_3b_3 Mabcdef etc show stereoisomerism, for e.g., Mabcdef have 15 geometrical isomers & all are optically active so it has 15 pairs of enantiomers thus in totality it has 30 stereoisomers. (a, b, c, d etc. are monodentate ligands while AB is bidentate ligand)

32. How many pair of enantiomers exits for $Ma_2b_2c_2$?

(A) 0 (B) 2
(C) 1 (D) 4

Space for Rough work

33. Which of the following can show geometrical as well as optical isomerism?

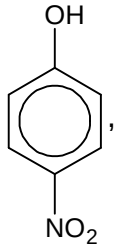
(A) $M(AB)_3$ (B) Mabcd (tetrahedral)

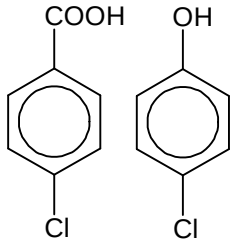
(C) Ma_3b_3 (D) Ma_4b_2

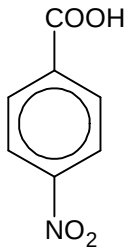
SECTION – C
(Single digit integer type)

This section contains **TEN** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

34. How many of the following are more acidic than HCOOH?

Picric acid, CH_3COOH, 

, HCl, 

HCN, H_2CO_3, 

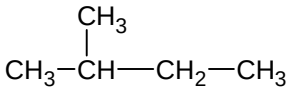
35. In how many of the following cases more than four atoms are in same plane

$XeF_4, PCl_5, \left[Ag(CN)_2\right]^-, \left[Pt(CN)_4\right]^{2-}, CH_3 - CH = CH - C \equiv C - CH_3$

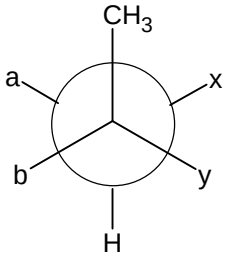
$\left[CO(NH_3)_6\right]^{3+}, CH_2 = C = CH_2, [PtCl_6]^{2-}$

Space for Rough work

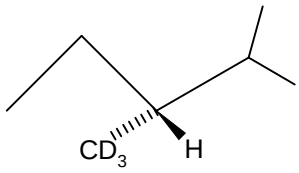
36. If EAN of metal in $K_4\left[M(CN)_6\right]$ is 36, find the number of d – electrons in complex.

37. 

If the Newmann projection of above compound can be written as mentioned below



How many combinations for a, b respectively are possible?

38. 

Monochlorination → how many enantiomeric pairs can be formed.

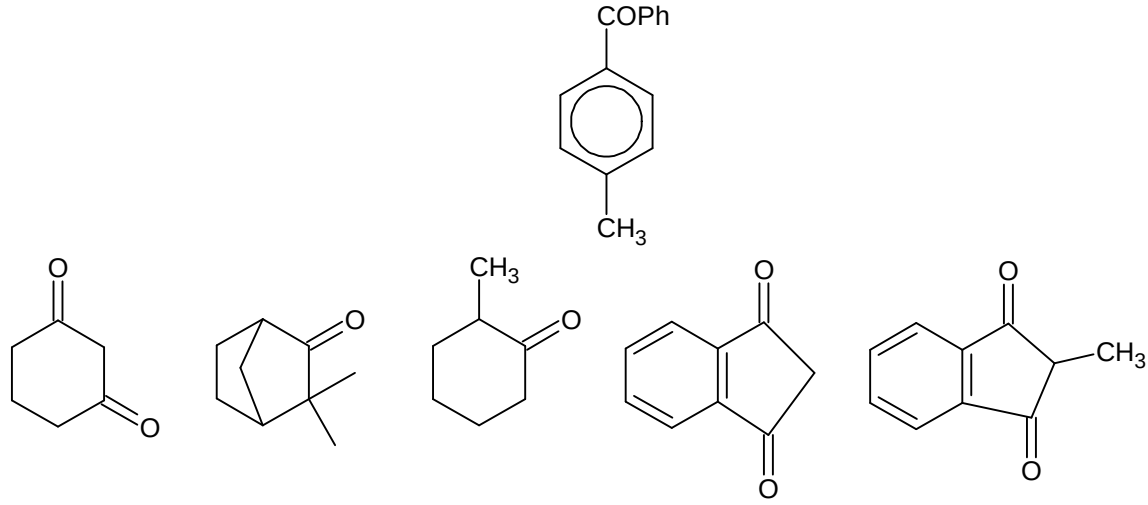
Optically pure

39. $H_2O_2 \longrightarrow H_2O + O_2$ the rate const. of this reaction is $2\times10^{-2}\,min^{-1}$(at certain temp.) & reaction is started with 0.5 M conc. of H_2O_2, after 23.03 min the reaction was suddenly stopped by lowering the temp & the remaining H_2O_2 was completely reacted with NaOCl, If no. of moles of O_2 produced in second reaction is $\frac{1}{a}$ find a. (log1.5 = 0.2)

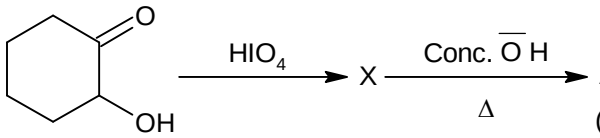
Space for Rough work

40. A 8 gm sample of iron are containing V%(by wt) of iron $\left(Fe^{2+}\right)$ was present in 1 L solution. This solution required V ml of a $KMnO_4$ solution for oxidation of Fe^{2+} in acidic medium. If in another titration 20 ml of same $KMnO_4$ is required, for oxidation of 20 ml H_2O_2 in acidic medium. Find volume strength of H_2O_2

41. How many of the following can give yellow ppt. with I_2 / OH^- ?



42.



(A contains 3 oxygen atom per molecule)

$$A \xrightarrow[\Delta]{\text{acidified with } H^+} C$$

DBE of C is

43. How many of the compounds are more acidic than phenol?
O – cresol, m – cresol, p – cresol, o – fluorophenol, o – chlorophenol,

o – bromophenol

Space for Rough work

SECTION – D
(Numerical Based XXXXX.XX answer Type)

This section contains **3 questions**. Each question, when worked out will result in numerical answer Type with answer xxxxx.xx.

44. A hydrocarbon with lowest molecular weight has only one chiral carbon. Find its molecular weight.

45. Electrolysis of aq. Solution of CH_3COOK is carried out. If 2 amp current is passed for 402 minutes, what volume (at STP) of gases could be produced if current efficiency is 80% (assume no absorption of gases)

46. An impure sample of As_2O_3 weighing 20 gm was dissolved in water containing 10 gm of $NaHCO_3$ and the final solution was diluted to 400 ml. 20 ml of this solution was oxidised by 25 ml of a solution of I_2. Another solution of hypo containing 1.24 gm of $Na_2S_2O_3.5H_2O$ in 40 ml was used to exactly reduce 25 ml of same I_2 solution (impurities inert) calculate % of AS_2O_3 in the sample (at. wt. of As = 75)

Space for Rough work

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **3 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D) out of which **ONLY ONE** option is correct.

47. If tangent is drawn at (x_1, y_1) on the hyperbola $\frac{x^2}{4}-\frac{y^2}{3}=1$ intersect the x-axis at C $(\alpha, 0)$ and y-axis at D $(0, \beta)$, then $\int_2^{4\sqrt{2}} \frac{\alpha}{\beta} dx_1$ is

(A) $\frac{3\pi}{8}$ (B) $\frac{-\pi}{4}$

(C) $\frac{-\pi}{8}$ (D) $\frac{\pi}{8}$

48. If $\int_0^x \frac{(t^3+t)dt}{(1+3t^2)} = f(x)$, then $\int_0^1 f'(x)dx$ is equal to

(A) $\frac{1}{6}-\frac{1}{9}\ln 3$ (B) $\frac{1}{6}-\frac{1}{9}\ln 4$

(C) $\frac{1}{6}+\frac{1}{9}\ln 3$ (D) $\frac{1}{6}+\frac{1}{9}\ln 4$

49. If $y=f(x)$ is solution of differential equation $\frac{dy}{dx}=\frac{y}{x}\frac{(1-3x^2y^3)}{(2x^2y^3+1)}$, then $f^3(x).x^3$ is equal to

(A) $x+cy$ (B) $2x+cy$

(C) x^2+cy (D) x^3+cy^2

Space for Rough work

(One or More than one correct type)

This section contains **FIVE** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

50. If $5x\left(1+\frac{1}{x^2+y^2}\right)=12$ and $5y\left(1-\frac{1}{x^2+y^2}\right)=4 \quad \forall x,y \in R$, then correct option is/are

(A) $|x|+|y|=3$ (B) $x+y=\frac{1}{5}$

(C) $|x|+|y|=\frac{3}{5}$ (D) $x^2+y^2=5$

51. If a tangent on ellipse at A (1, 1) intersect its directrix at B (7, –6) and S be the focus of ellipse and C (α, β) is the centre of ΔSAB, then

(A) $\alpha+\beta=1$ (B) $\alpha-\beta=7$

(C) $SC^2=20.5$ (D) $SC^2=21.25$

52. If $f(x)=\frac{\cos^{-1}\left(1-\{x\}\right).\sin^{-1}\left(1-\{x\}\right)}{\sqrt{2\{x\}}.\left(1-\{x\}\right)}$ here {.} denotes fractional part of x then;

(A) $\lim\limits_{x\to 0^+} f(x)=\sqrt{2}\lim\limits_{x\to 0^-} f(x)$ (B) $\lim\limits_{x\to 0^-} f(x)=\sqrt{2}\lim\limits_{x\to 0^+} f(x)$

(C) $\lim\limits_{x\to 0^-} f(x)=\frac{\pi}{2\sqrt{2}}$ (D) $\lim\limits_{x\to 0^-} f(x)=\pi\sqrt{2}$

53. If $\int_a^b |\sin x|dx=8$ & $\int_0^{a+b} |\cos x|dx=9$ then;

(A) $a+b=9\pi$ (B) $b-a=4\pi$

(C) $\int_a^b x\sin x dx=2\sqrt{2}\pi$ (D) $\int_a^b x\sin x dx=-2\sqrt{2}\pi$

54. If a, b, c are sides of an acute angled triangle satisfying $a^2+b^2+c^2=6$ then $(ab+bc+ca)$ can be equal to;

(A) 3 (B) 4

(C) 5 (D) 7

Space for Rough work

(Paragraph Type)

This section contains **ONE** paragraph. Based on the paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Paragraph for Question Nos. 55 and 56

If locus of point $P(z)$ in complex plane is $|z+z_1|+|z+z_2|=4$, where A represents z_1 as (1, 0) and B represents z_2 as (–1, 0) and $Q(\omega)$ is moving point inside the locus of $P(z)$ such that all internal angle bisectors of triangle ΔPAB concurrent at Q. Then, answer the following questions if $|\omega-\omega_1|+|\omega-\omega_2|=2$.

55. $|\omega_1|+|\omega_2|$ is equal to

(A) $\frac{2}{\sqrt{3}}$ (B) $\sqrt{\frac{2}{3}}$

(C) $2\sqrt{\frac{2}{3}}$ (D) $\frac{2\sqrt{2}}{3}$

56. If minimum value of $|\omega-z_1|+|\omega-z_2|$ is equal to m, then $[m]$ is (where [.] denotes greatest integer part)

(A) 1 (B) 2

(C) 3 (D) 4

SECTION – C
(Single digit integer type)

This section contains **TEN** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

57. If $f(2+x)+f(x)=2$ for all $x\in[0,2]$ and $k=\int_0^4 f(x)dx-4$ such that both roots of $ax^2-bx+c=0$ lies between k & q where $q=\frac{\alpha}{\beta}$ such that $\alpha+\beta=1$ & $\alpha^2+\beta^2=\frac{1}{2}$ then least positive integral value of a_____.

Space for Rough work

58. There are "n" married couples at a party each parson shakes hand with every person other than her or his spouse then number of handshakes is $f(n)$, $\left(\sum_{n=4}^{\infty}\frac{1}{f(n)}\right)^{-1} = ?$

59. If $f(x) = x + \int_0^1 x^2.zf(z)dz + \int_0^1 xz^2(z)dz$ then, we have $\lim_{x\to\infty}\frac{f(x)}{\left(ax^2+7x\right)} = \frac{20}{119}$ then "a" = ?

60. If x is root of $\left(1+x^4\right) = 7(1+x)^4$ then $\left|x+\frac{1}{x}\right| = ?$

61. If y = 2 is directrix and (0, 1) be the vertex of parabola $x^2 + \lambda y + \mu = 0$ then; $(\lambda - \mu) = ?$

62. The minimum number of real roots of $\left(x^2+3x+a\right)\left(x^2+ax+1\right) = 0$ equals

63. If z_1, z_2, z_3 are three complex numbers, such that $|z_1| = |z_2| = |z_3| = 1 \& z_1^2 + z_2^2 + z_3^2 = 0$ then $\left|z_1^3 + z_2^3 + z_3^3\right|$ is equal to_____. (not equal to 1)

64. A variable plane is at constant distance 3 from origin and meets the axis in X, Y, Z planes through X, Y, Z parallel to coordinate planes are made. If P be the point of intersection of above made planes, given $P(\alpha,\beta,r)$ then $\frac{\alpha\beta r}{\sqrt{\alpha^2\beta^2+\beta^2r^2+r^2\alpha^2}} = ?$

65. If $\left[\bar{a}\ \ \bar{b}\ \ \bar{c}\right] = 2$ and $\left(\vec{a}.\vec{b}\right) = 2$, $\left(\vec{c}.\vec{a}\right) = 5$, then $\vec{a}$ is equal to $x\bar{b} + y\bar{c} + z\left(\bar{b}\times\bar{c}\right)$ (where $\bar{b}$ & $\bar{c}$ are two non-coplanar orthogonal unit vectors) then (x + y – z) = ?

66. If $\cos^{-1}\left(2-x^2\right) + \sin^{-1}\left(2-x^2\right) + \tan^{-1}x \geq \frac{2\pi}{3}$ have no solution if $x < \alpha$ or $x > \beta$ $(\text{for } \alpha < \beta)$, then $\left\{\frac{1}{\max(\alpha)} + \left(\min(\beta)\right)^2\right\} = ...$

Space for Rough work

SECTION – D
(Numerical Based XXXXX.XX answer Type)

This section contains **3 questions**. Each question, when worked out will result in numerical answer Type with answer xxxxx.xx.

67. If $f(x)=\frac{3x^2+1}{\sqrt{x^4+x^2}}\ \forall 0<x\le\sqrt{2}$, then $f(x)$ has minimum value M at x = m. Then, $\left(m^2+\frac{1}{M^2}\right)$ is equal to

68. If $x=\int_0^{\infty}\frac{dt}{(1+t^2)(1+t^{2017})}$, then $\frac{3x}{\pi}$ is equal to

69. If $\log_{(3x+4)}(4x^2+x+1)+\log_{(2x+1)}(6x^2+11x+4)$ then x is _____.

Space for Rough work